

Class IX

Python Programming Language

Theoretical Questions

1. What is Python?

Python is a high-level, general-purpose programming language. Its design philosophy emphasizes code readability with the use of significant indentation.

2. In which area Python Programming is useful for?

Python Programming is useful in the area of:

- web development (server-side),
- software development,
- mathematics,
- system scripting.

3. What can Python do?

Python can be used on a server to create web applications.

- Python can be used alongside software to create workflows.
- Python can connect to database systems. It can also read and modify files.
- Python can be used to handle big data and perform complex mathematics.
- Python can be used for rapid prototyping, or for production-ready software development.

4. Why Python?

Python is preferred programming language due to the following reasons:

- Easy to learn: It has fewer keywords and simple structure of instructions, so less efforts to learn Python.
- Free and open source
- Portable: A Python program can run on wide variety of hardware platforms.
- Extensive library support: Python saves times and effort using the support of extensive library.

As an example, YouTube has been largely written in Python. Google makes extensive use of Python. NASA uses Python for scientific programming tasks.

5. What are the two modes of Python?

The two modes of Python are interactive and script mode.

6. What is interactive mode. List it's advantages and drawbacks.

As the name suggests, Interactive mode allows us to interact with OS. When we type Python statement, interpreter displays the result(s) immediately.

Advantages:

- Python, in interactive mode, is good enough to learn, experiment or explore.
- Working in interactive mode is convenient for beginners and for testing small pieces of code.

Drawback:

- We cannot save the statements and have to retype all the statements once again to re-run them.

7. What is script mode. List its advantages and drawbacks.

In script mode, we type python program in a file and then use interpreter to execute the content of the file. Scripts can be saved to disk for future use. Python scripts have the extension .py, meaning that the filename ends with .py.

8. What are variables and how to create the same in Python?

Variables – Variables are containers for storing data values.
Creating Variables – Python has no command for declaring a variable.

A variable is created the moment you first assign a value to it. Variables do not need to be declared with any type and can even change type after they have been set.

For example:

```
x = 5  
y = "John"
```

Here x is variable which is assigned a value 5 which is automatically declared as integer as data type and y is variable which is assigned a value 'John' which is automatically declared as string data type.

9. Discuss Data Types in Python?

Python supports a number of data types such as number, string, list, tuple, set and dictionary. We will discuss about number and string data type here:

Number: The following are the two important number data type:

a. int – int stands for integer which is a positive or a negative number without a decimal point.

There are three types of integers in Python:

Example: a = 7; b=-90

b. float – float or floating-point number is positive or negative number with a fractional part.

Example: c = 9.7; d = -88.9

Strings: Strings in Python are a sequence of characters. These characters are enclosed within single quotes ('), double quotes (") or triple quotes ("). Triple quotes are used when the string value extends over more than one line.

Example of strings:

```
name = "Myra"
```

```
section = 'A'
```

```
message = """A dream does not become reality through magic; it takes sweat, determination, and hard  
work. ~ Colin Powell"""
```

10. What are operators in Python?

Operators are special symbols that are used to perform operations on variables and values. Operators are used in everyday Mathematics, like +, -, x, ÷ to perform addition, subtraction, multiplication, and division of values. Python also has a number of operators that help us perform operations of various types.

11. What is assignment operator?

Assignment operator (=) is operator which is used to assign a value to a variable.

For example:

```
x = 5  
y = "John"
```

12. What are arithmetic operators in Python? Explain with an example.

Arithmetic operators are operators used to perform mathematical operation on variables and value.

For example, in 5 + 9, '+' is arithmetic operator performing addition on values 5 and 9.

Similarly, in p / q, '/' is arithmetic operator performing division on p and q.